GRADE 7 PHYSICS – CHAPTER 2

States of Matter, Density, Pressure, Hydrostatic Pressure, and Pascal's Principle

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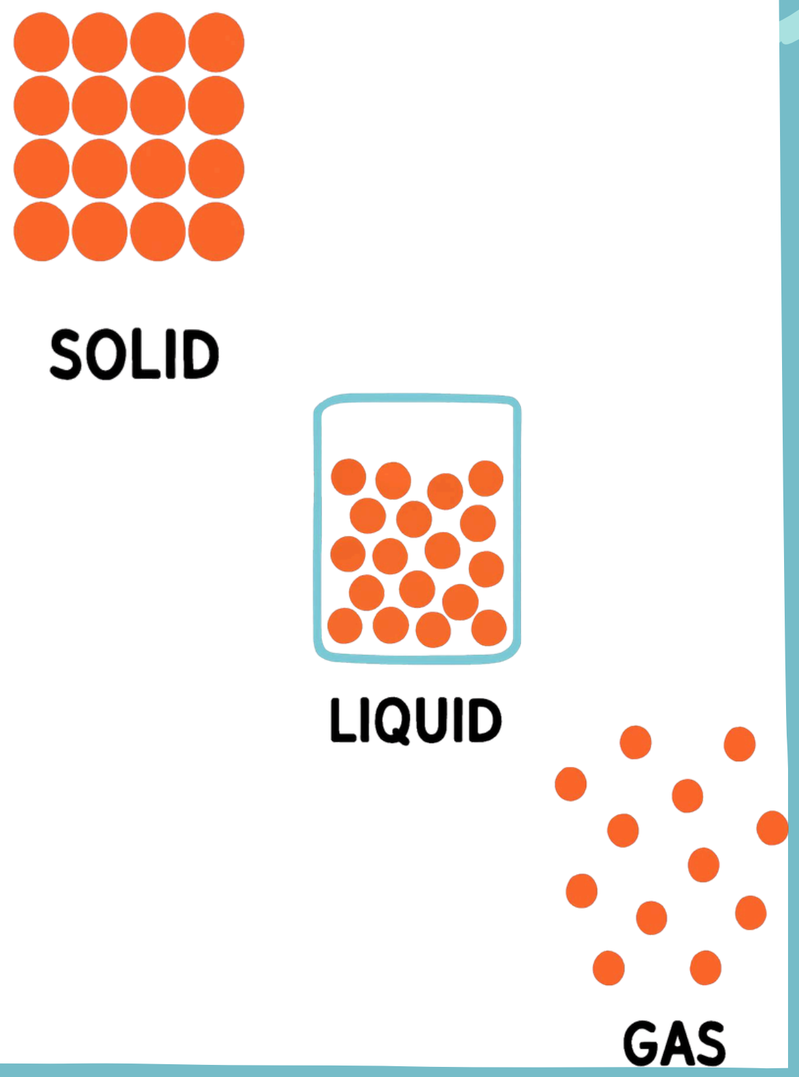
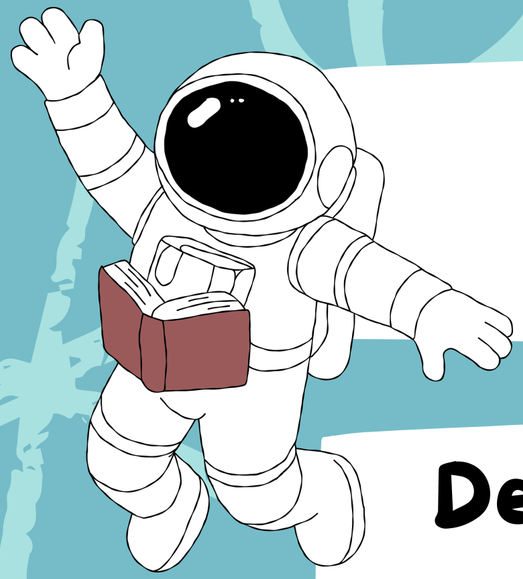
Solid, Liquid, Gases

Solids, liquids, and gases are the three main states of matter. They differ in how their particles are arranged and how they move.

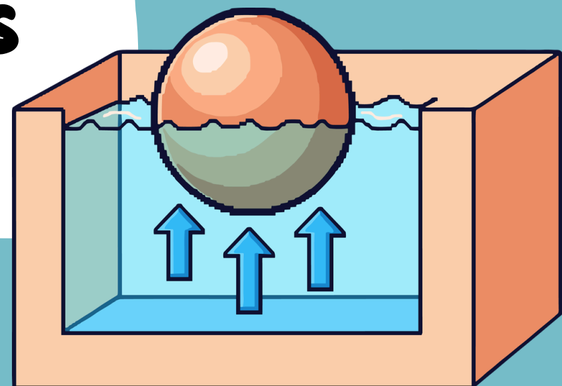
- Solid: definite shape and volume. Particles tightly packed and only vibrate.
- Liquid: fixed volume but shape follows container. Particles move past each other.
- Gas: no fixed shape or volume. Particles far apart and move freely.





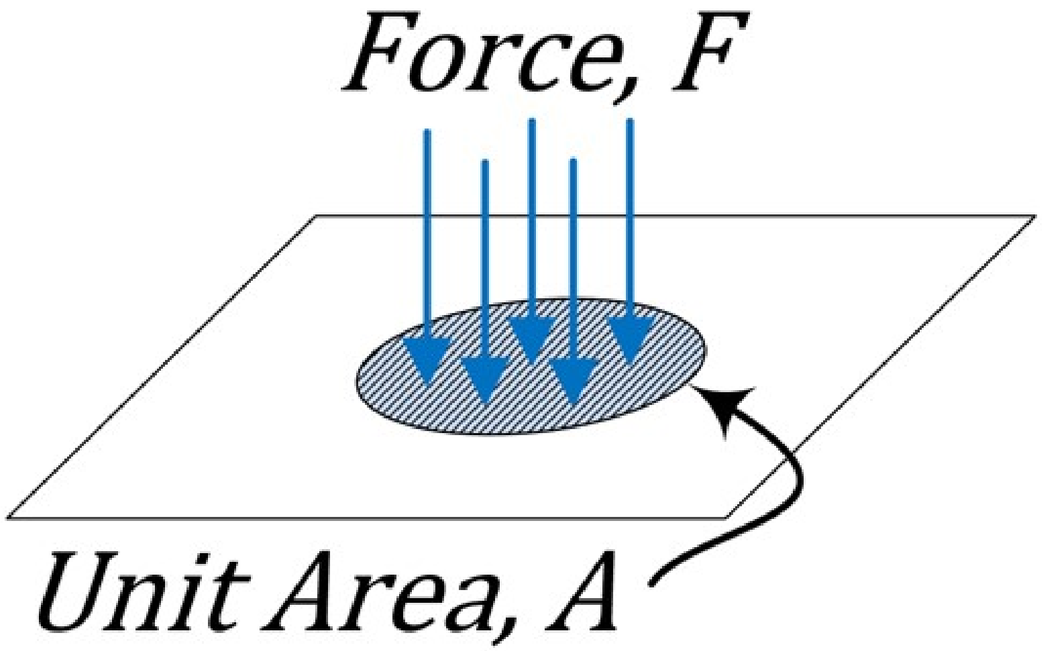
Density

Density is the mass per unit volume of a substance. It can be calculated using the formula $\rho = m / V$, where ρ is density, m is mass, and V is volume. Density helps explain why objects float or sink in fluids.



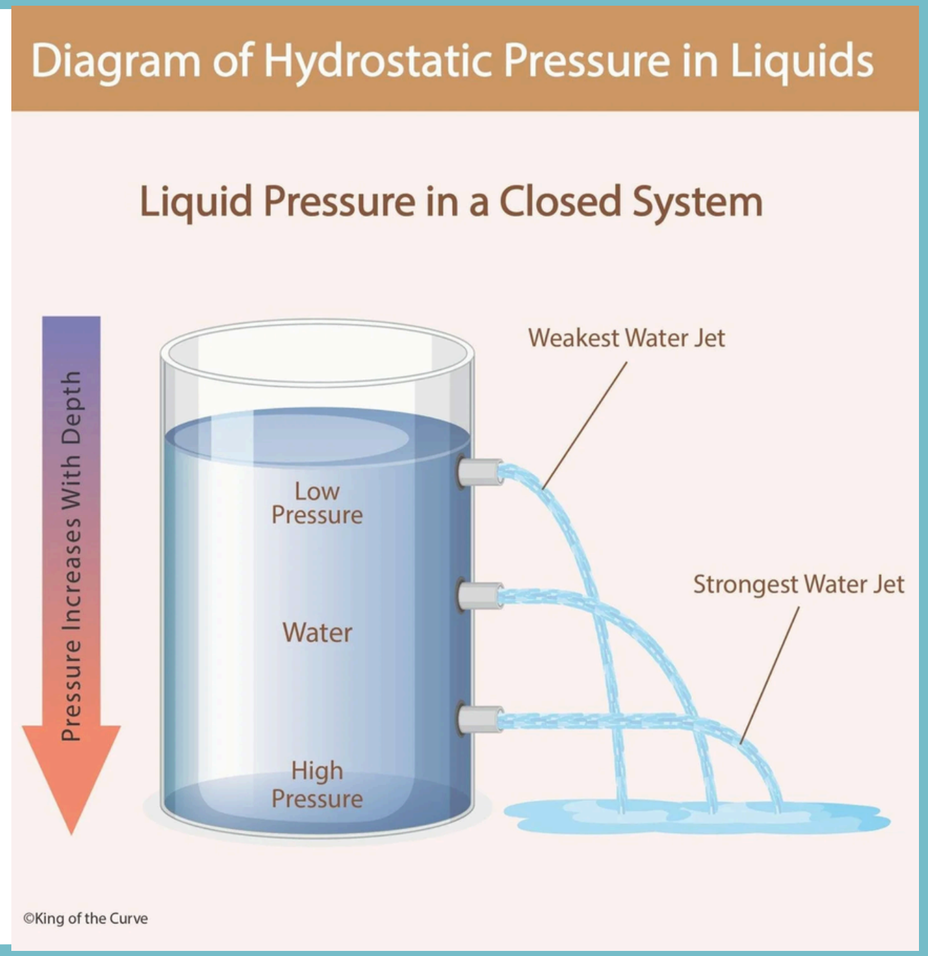
Pressure

Pressure is the force applied per unit area. It is calculated using the formula P = F / A, where P is pressure, F is force, and A is area. Pressure increases when the force becomes larger or when the area becomes smaller.



Hydrostatic Pressure

Hydrostatic pressure is the pressure in a fluid at rest. It can be calculated using the formula $P = \rho gh$, where ρ is density, g is gravity, and h is depth. The pressure increases as the depth increases.



Pascal Principle

Pascal's Principle states that pressure applied to an enclosed fluid is transmitted equally in all directions. This principle is used in hydraulic systems to multiply force, such as hydraulic jacks and lifts.

Thank you